



Chapter 1 Java Fundamentals



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Content

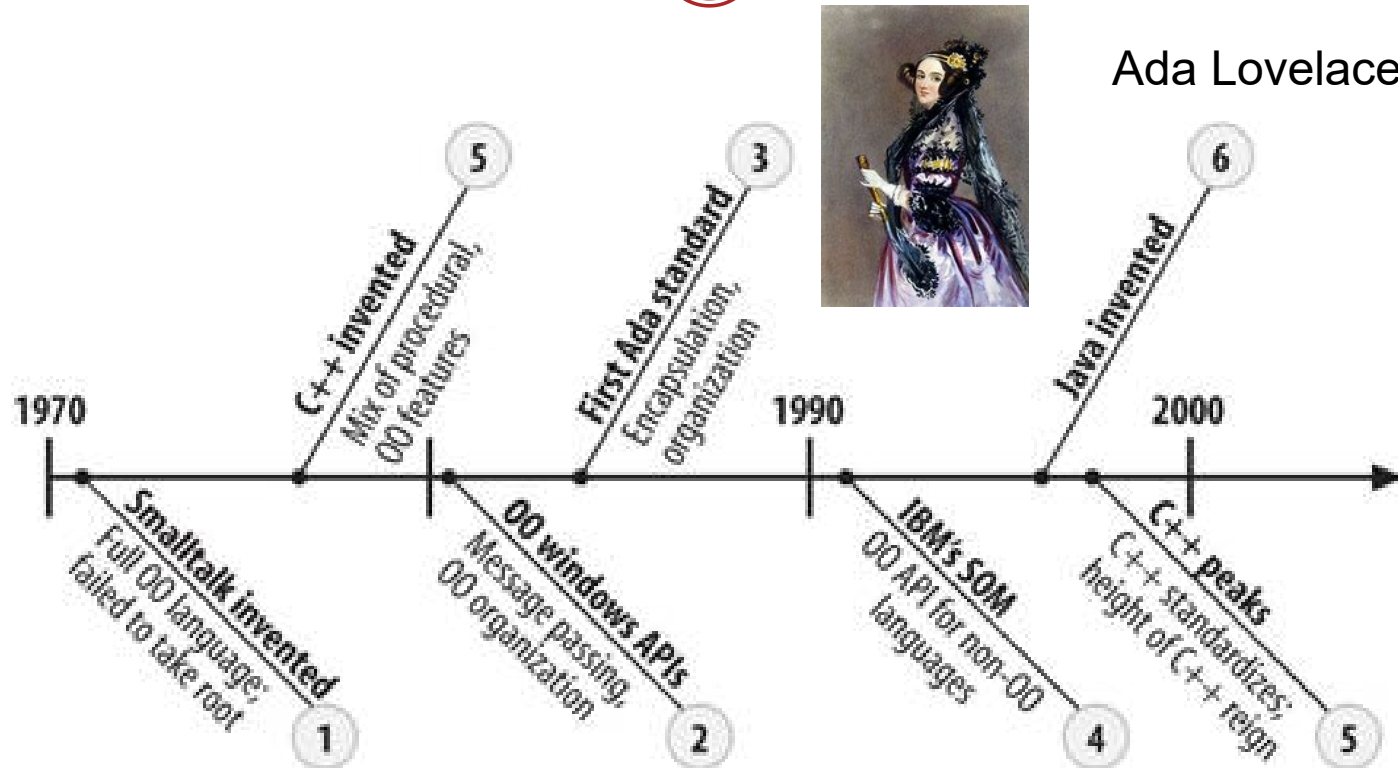
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- Evolution of Java
- JDK and JRE
- Java Operating Mechanism
- Java Developing Environment
- Java Primary Data Types
- Java Basic Grammar



Evolution of Java – Success of OOP

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from 《Beyond Java》



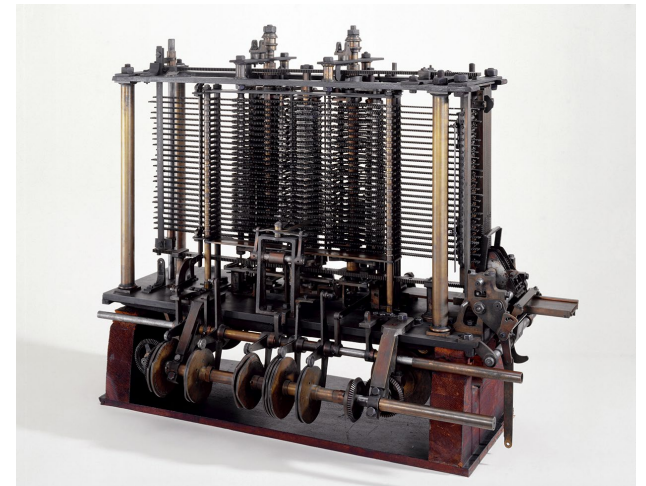
Ada Lovelace (历史上第一位程序媛)

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- 10 December 1815 – 27 November 1852)
- English mathematician and writer
- The first to recognize that the machine had applications beyond pure calculation
- She published the first algorithm



mechanical general-purpose computer,
the Analytical Engine





Evolution of Java – Life of Java

Open Discussion:
Please list some
resource-limited
devices

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- Past

- Resource-limited Device
- C++
- Green Project
- Oak
- Mosaic / Netscape / Mark Andreessen
- HotJava

- Present

- Internet / WWW
- Enterprise
- 1st language in industry

- Future

- Java vs. Dynamic Language
- Java and open source

James Gosling: Dr.Java

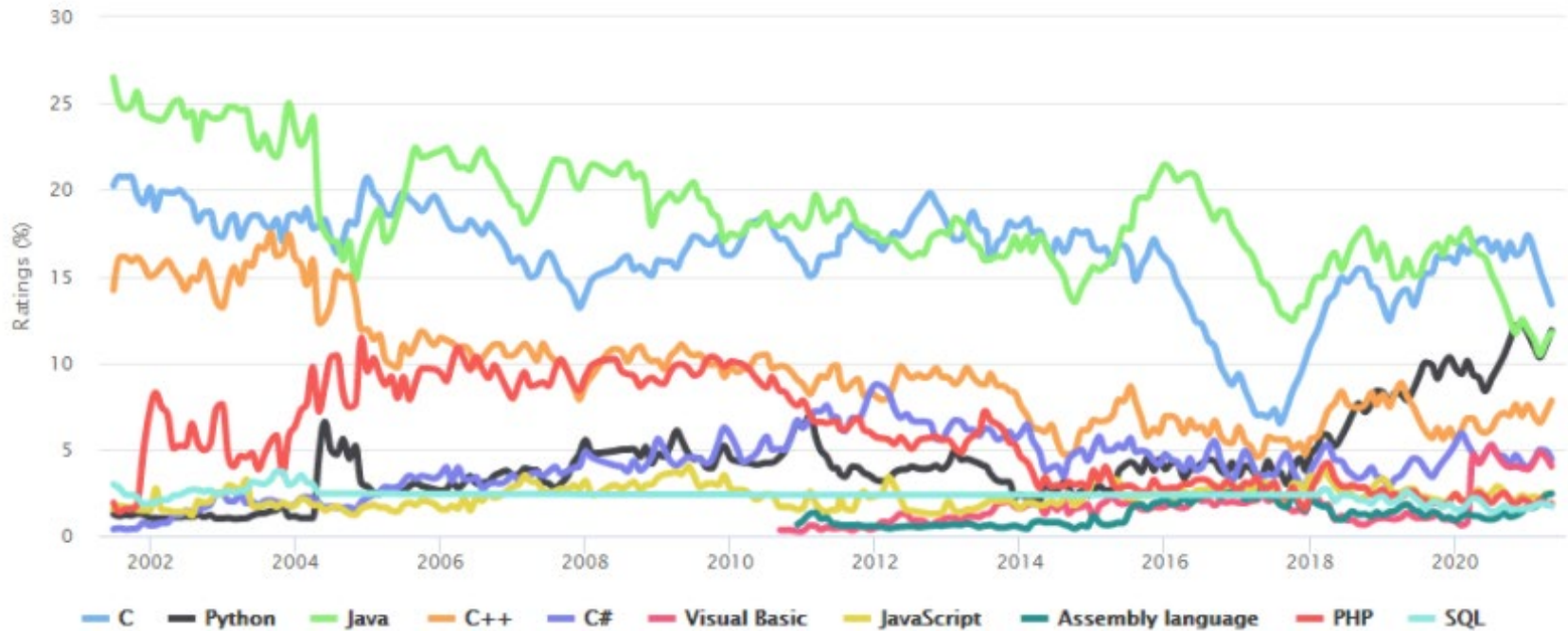




Today's Java

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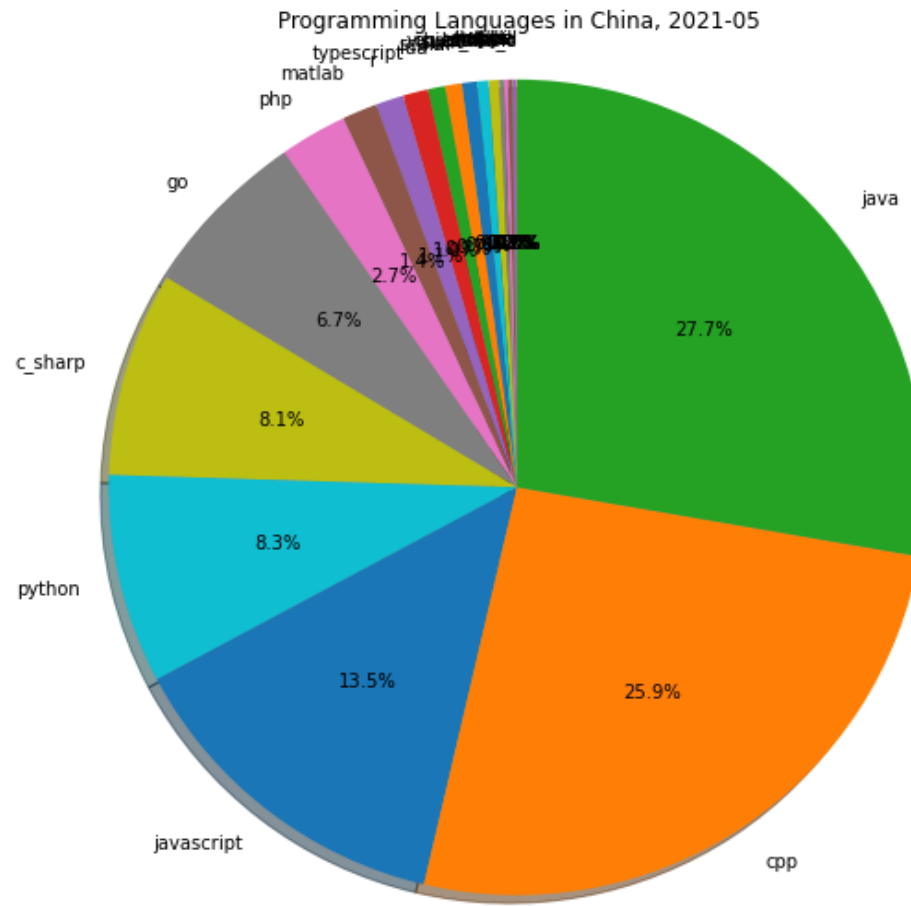
TOP 10编程语言TIOBE指数走势图 (2002-2020)





Today's Java

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Today's Java

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Salary:

rank	pl_	salary_mean	salary_median	salary_95_min	salary_95_max	head_count	percentage
1	rust	24293	24000	7000	45000	1022	0.23%
2	scala	20041	17500	7000	45000	4158	0.93%
3	julia	19815	20000	12500	27500	54	0.01%
4	python	19329	17500	6500	45000	51652	11.58%
5	go	19246	17500	7000	45000	42051	9.43%
6	lua	19180	17500	6788	45000	4192	0.94%
7	perl	18972	17500	7000	40000	2816	0.63%
8	matlab	18905	17500	7000	37500	8629	1.93%
9	swift	18205	16000	7000	40000	3719	0.83%
10	r	17547	17500	6000	40000	6904	1.55%
11	ruby	17021	15000	5250	37500	1211	0.27%
12	kotlin	16630	15000	5750	40000	2601	0.58%
13	c/c++	16597	15000	6000	37500	161715	36.26%
14	haskell	16250	16250	15000	22500	6	0.00%
15	typescript	16210	15000	7000	36842	6239	1.40%
16	java	15256	13000	5500	37500	172709	38.72%



JDK and JRE

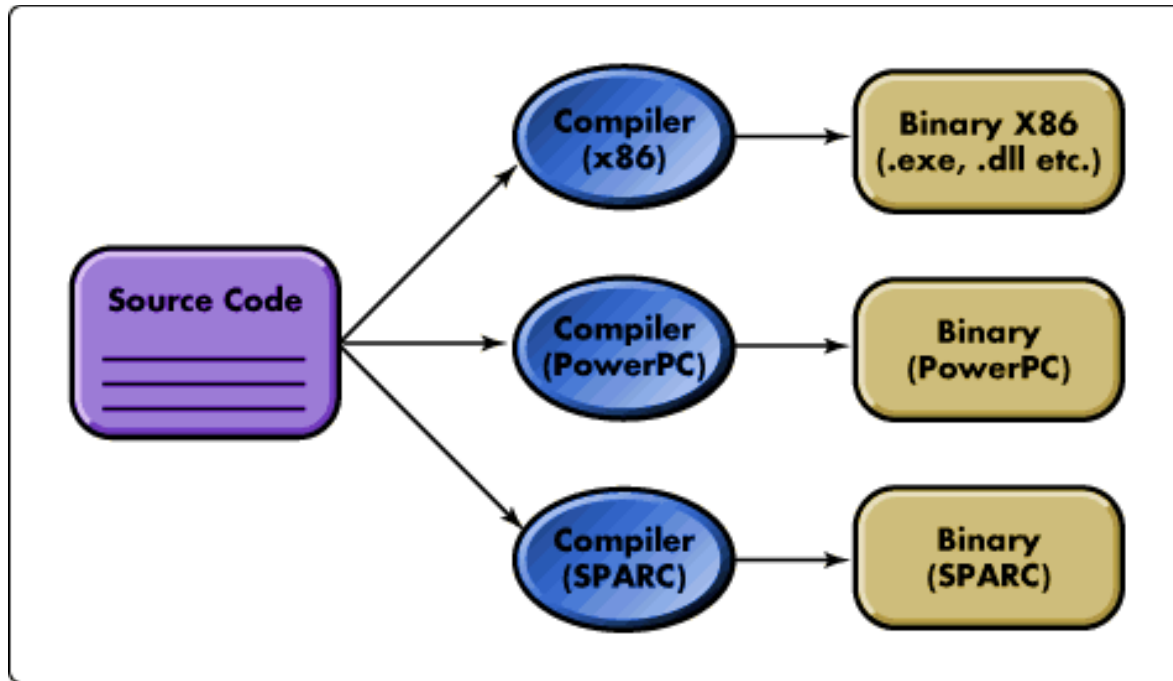
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- JDK – Java Development Toolkit
 - J2SE – Java 2 Standard Edition
 - J2EE – Java 2 Enterprise Edition
 - J2ME – Java 2 Micro Edition
- JRE – Java Runtime Environment



Java Mechanism – Traditional

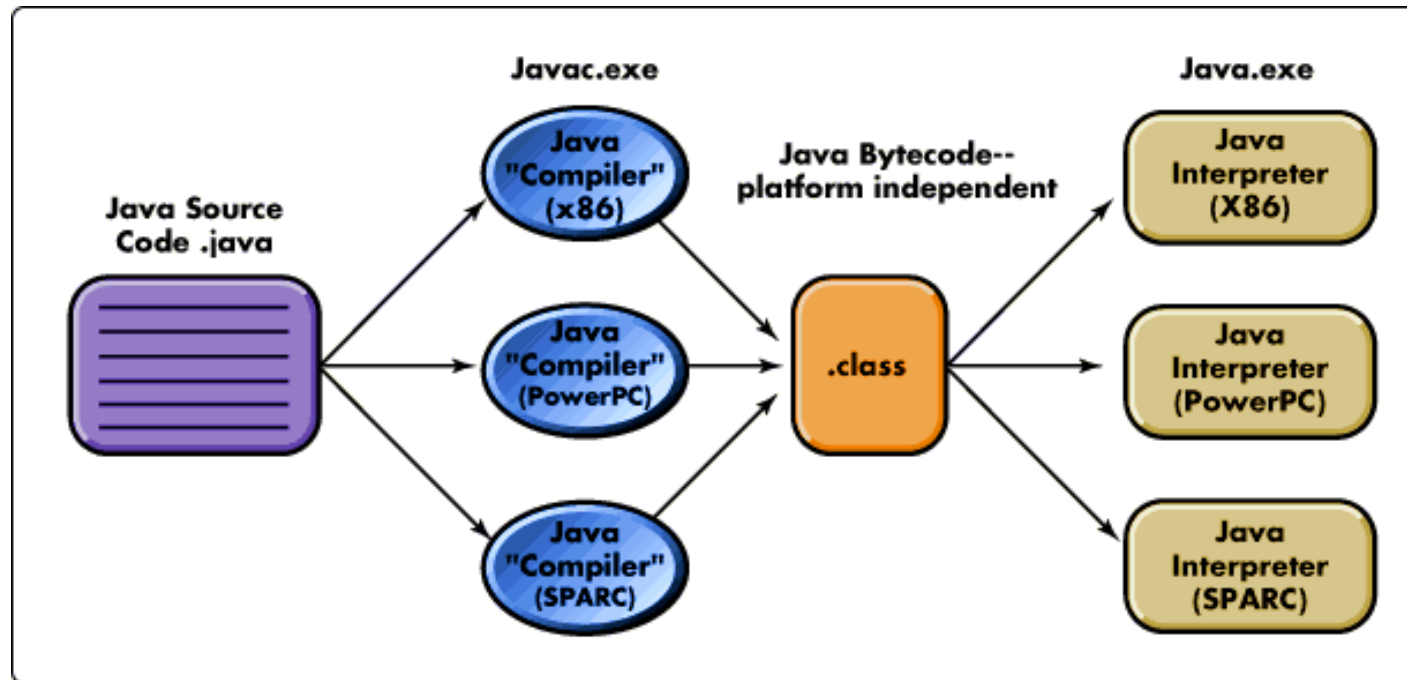
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Java Mechanism – Java

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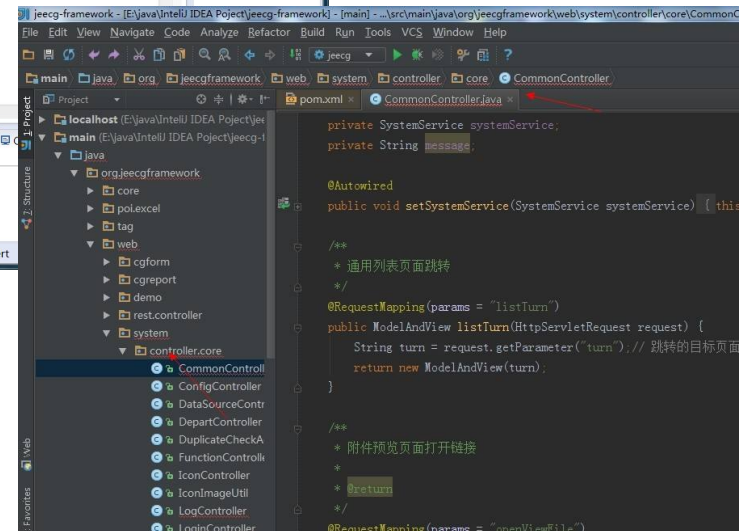
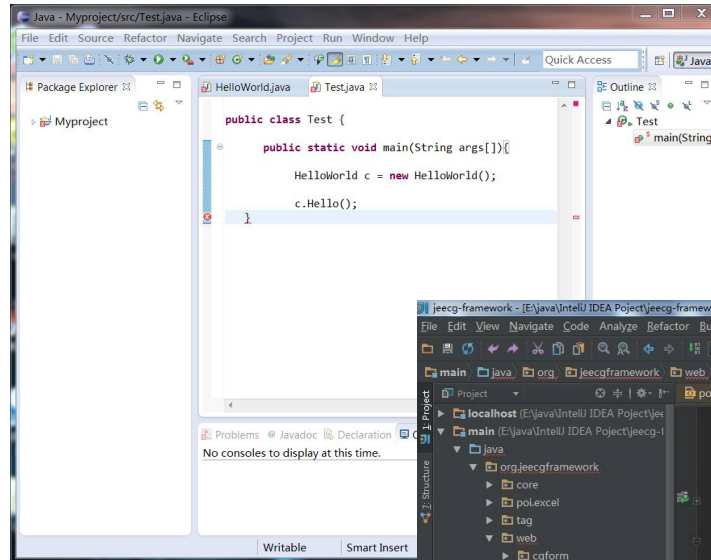




Java Developing Environment

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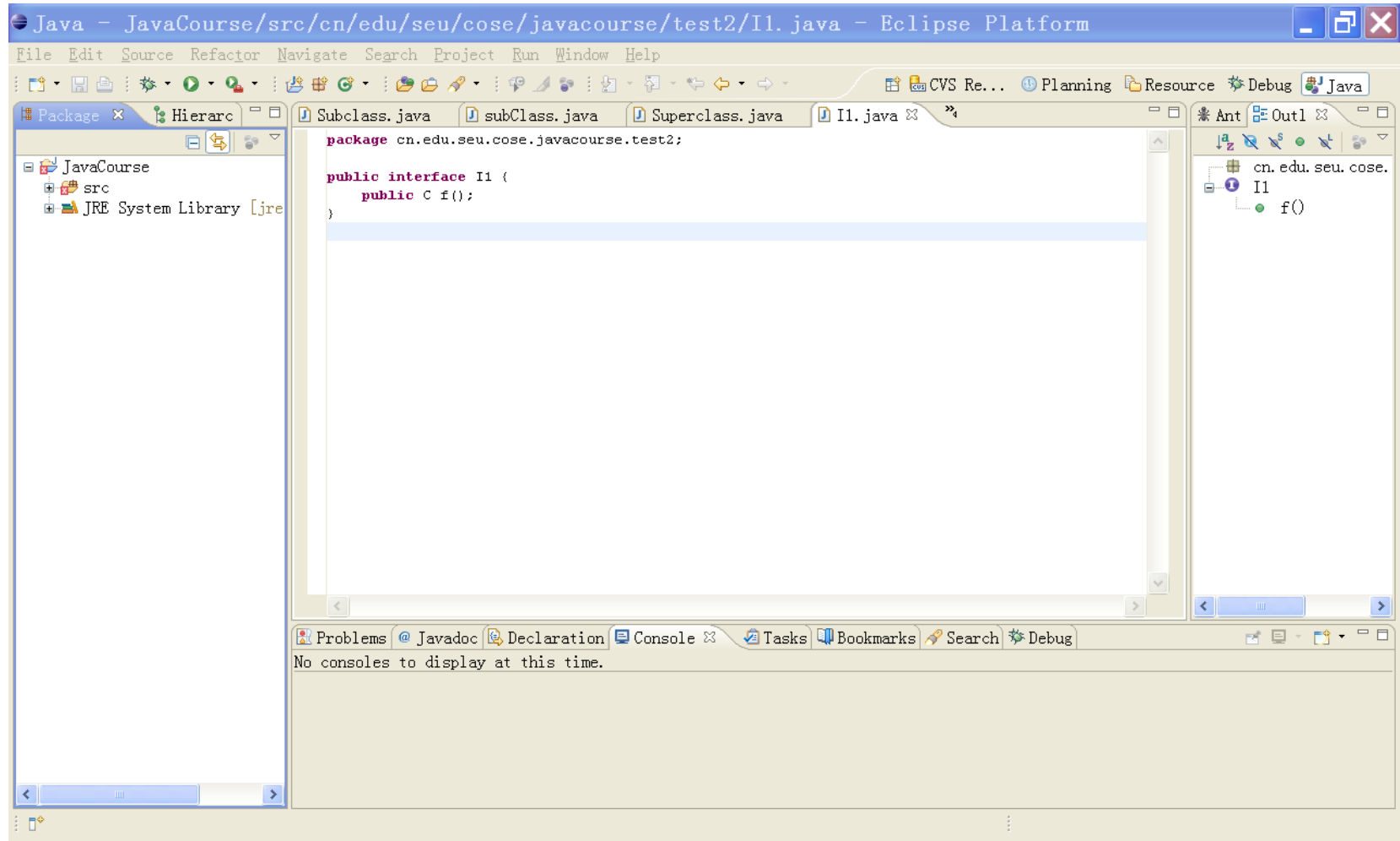
- Text editor
- IDE
 - Eclipse
 - IntelliJ IDEA
 - Netbeans
 - MyEclipse





Eclipse

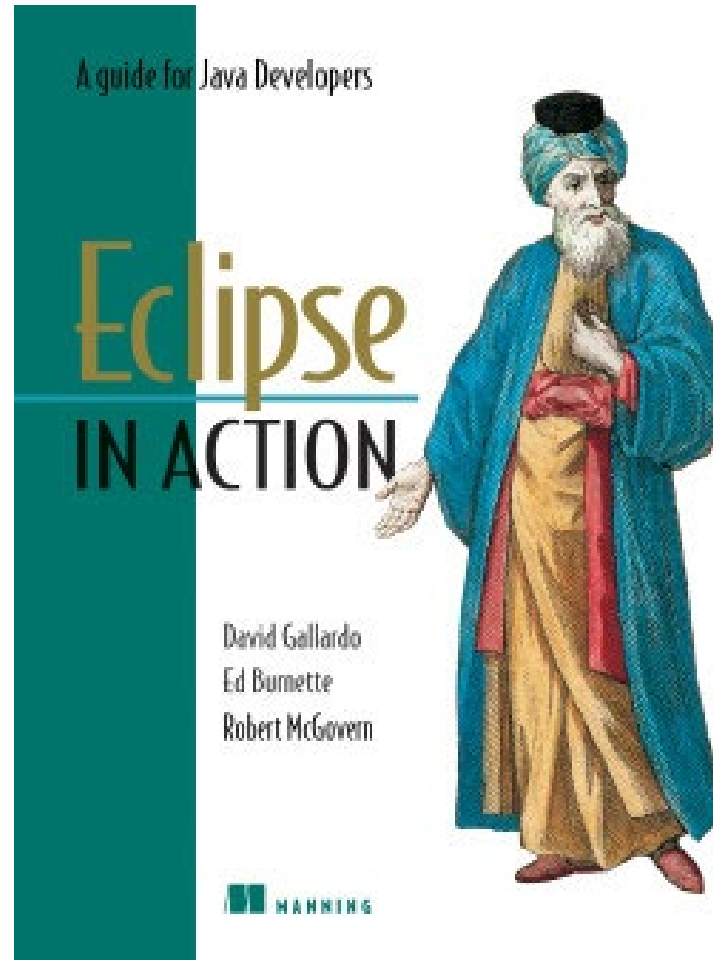
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Eclipse is not only an IDE

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Java Features

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- Simplicity: simple grammar, rich library
- Pure OO: everything is object!
- Security: memory access, garbage collection, exception
- Portability: Java Virtual Machine
- Interpreted execution: Bytecode



Exploring Java

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```
package cn.edu.seu.cose.javacourse.ch01;
public class Person {
    private String name;
    private int age;
    public Person(String name, int age){
        this.name = name;
        this.age = age;
    }
    public void greet(){
        System.out.println("Hello, I am " + name
            + " , and I am " + age + " years old");
    }
    public static void main(String[] args){
        Person tom = new Person("Tom", 18);
        tom.greet();
    }
}
```




The Structure Of Java Programs

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package declaration	←-----	<code>package</code> cn.edu.seu.cose.javacourse.ch01;
class declaration	←-----	<code>public class</code> Person {
variable declaration and initialization	←-----	<code>private String</code> name;
		<code>private int</code> age;
constructor	←-----	<code>public</code> Person(<code>String</code> name, <code>int</code> age){
		<code>this.name</code> = name;
		<code>this.age</code> = age;
		}
method	←-----	<code>public void</code> greet(){
		<code>System.out.println</code> ("Hello, I am " + name
		+ " , and I am " + age + " years old");
		}
main method	←-----	<code>public static void</code> main(<code>String</code> [] args){
		Person tom = <code>new</code> Person("Tom", 18);
		tom.greet();
		}
		}

```
public class Person {
```

How many errors?

```
    privat String name;
```

```
    privat int age;
```

```
    System.out.println("the program begins.");
```

```
    public void person(int age){
```

```
        this.age = age
```

```
    }
```

```
    public int greet{
```

```
        System.out.println("Hello, I am Tom, and I am "  
            + age + " years old");
```

```
    }
```

```
    private static main(String arg){
```

```
        Person tom = new Person("18");
```

```
        tom.greet();
```

```
    }
```

Java Primary Data Types

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Java Primary Data Types

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Open Discussion: Why we need wrappers when we already have primary type?
And why we need primary type when we have wrappers?

Type	size(bit)	range	wrapper
boolean	1	true/false	Boolean
char	16	Unicode	Character
byte	8	[-128, 127]	Byte
short	16	$[-2^{15}, 2^{15}-1]$	Short
int	32	$[-2^{31}, 2^{31}-1]$	Integer
long	64	$[-2^{63}, 2^{63}-1]$	Long
float	32	3.4×10^{38}	Float
double	64	1.7×10^{308}	Double
void			Void



Primary Types and Wrapper

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- Values of Primary Types are NOT Objects!
- Each Primary type has a corresponding wrapper to wrap a value into an object:
 - Integer a = 473;
 - System.out.println(a.compareTo(new Integer(472)));
 - int a = 473;
 - System.out.println(a.compareTo(472));





Conversion Between Values

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- From Low Accuracy to High Accuracy: Auto
 - `double d = 10;`
- From High Accuracy to Low Accuracy: Cast
 - `int t = (int)10.2;`



More About This Statement

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Class:java.lang.System method,void method,int

```
System.out.println(a.compareTo(new Integer(472)));
```

object:PrintStream, static

object:Integer



Print and Format

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- `System.out.println()`
- String Formatter

```
double pi = 3.1415926;  
String result = String.format("%.2f", pi);  
System.out.println(result);  
// print pi with specific digits of fractional part
```




Variables and Constants

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- Declare and use
- Lifecycle and Hidden Variables

```
int a = 10;  
final int B = 20;
```

```
public class Test {  
    int t = 0;  
    public void hideT(){  
        int t = 10;  
        int s = 9;  
        System.out.println(t);  
    }  
    public void printT(){  
        System.out.println(t);  
    }  
}
```



Notice!

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- Different with C++



Duplicate local variable i

```
int i = 0;  
for(int j=0; j<10; j++){  
    int i = 10; // not allowed in java  
}
```

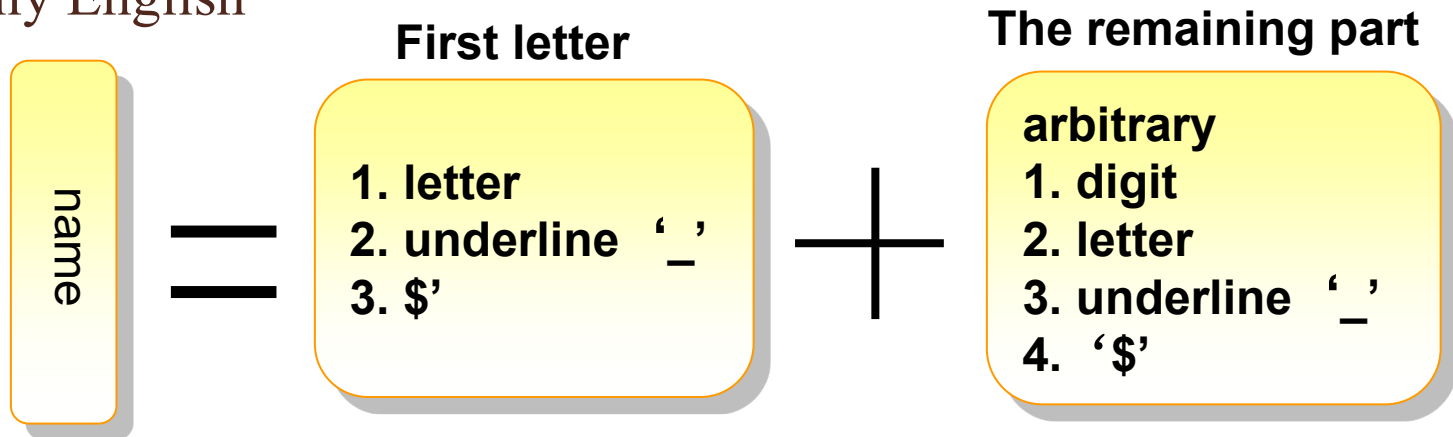


Naming

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- Basic Principle:

- A names should reflect the meaning of a class/package/variable...
- Different with Java keywords
- ****Different with java.lang.*** // not restricted
- Only English





Naming

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- project

demo

- package

package cn.edu...;

- class

public class Person;

- variable

int age = 20;

- method

void greet(){};

- constant

final double PI = 3.14;



Java Operator

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- Mathematical operator
- Relational operator
- Logical operator
- Bitwise operator
- Assignment operator
- Others



Mathematical Operator

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- $+$ 、 $-$ 、 $*$ 、 $/$ 、 $\%$
- $++$ 、 $--$



Relational Operator

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- > , >=

- < , <=

- == , !=

- instanceof

```
Person tom = new Person("Tom", 18);  
System.out.println(tom instanceof Person);
```



Logical Operator

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- `&`, `|`
- `&&`, `||`
- `!`
- `^`



Bitwise Operator

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- <<
- >>
- >>>



Assignment Operator

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- =
- += 、 -= 、 *= 、 /= 、 %=
- >>= 、 <<= 、 >>>=



Others

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- ? :

Ternary if-else operator

- .

```
return i < 10 ? i * 100 : i * 10;
```

- new

- []



Java Grammar

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- Package
- Import
- Class
- Field
- Method

```
package cn.edu.seu.cose.javacourse.test;
```

```
public class Person {
```

```
    private String name;
```

```
    private int age;
```

```
    public Person(String name, int age){
```

```
        this.name = name;
```

```
        this.age = age;
```

```
    }
```

```
    public void greet(){
```

```
        System.out.println("Hello, I am " + name  
                             + " , and I am " + age + " years old");
```

```
    }
```

```
    public static void main(String[] args){
```

```
        Person tom = new Person("Tom", 18);
```

```
        tom.greet();
```

```
    }
```

```
}
```



Lab Work 0

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- Write a Java Class Student
 - id, name, gender
 - A sample greet() outputs: “Hello, I am Xiang Zhang. I am a male student, and I am from Class 3.”
 - Open discussion:
 - ✦ Is it good to use a String for gender? Any better type?
 - ✦ Don't put everything in main(), why?



Java Statement

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- if-else
- switch
- while、 do-while
- for
- break
- continue
- return



Java Keywords

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abstract	else	interface	
assert	enum	long	
boolean	extends	native	switch
break	false	new	synchronized
byte	final	null	this
case	finally	package	throw
catch	float	private	throws
char	for	protected	transient
class	goto	public	true
const	if	return	try
continue	implements	short	void
default	import	static	volatile
do	instanceof	strictfp	while
double	int	super	



Java Comments

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```
// This is a simple lined comment
```

```
/* This is a multiple lined comment  
 * This is a multiple lined comment  
 * This is a multiple lined comment  
*/
```

```
/**  
 * @param age  
 * @return  
 */  
public int count(int age){  
    return 0;  
}
```




Lab Work 1

41

```
public class Person {  
  
    private String name;  
    private int age;  
  
    public Person(String name, int age) {  
        this.name = name;  
        this.age = age;  
    }  
  
    public void greet() {  
        System.out.println("Hello, I am " +  
            name + " , and I am " + age + " years old");  
    }  
  
    public static void main(String[] args) {  
        Person tom = new Person("Tom", 18);  
        tom.greet();  
    }  
}
```

bad code!
data is hard-coded, which is
hard to modify



Lab Work 1

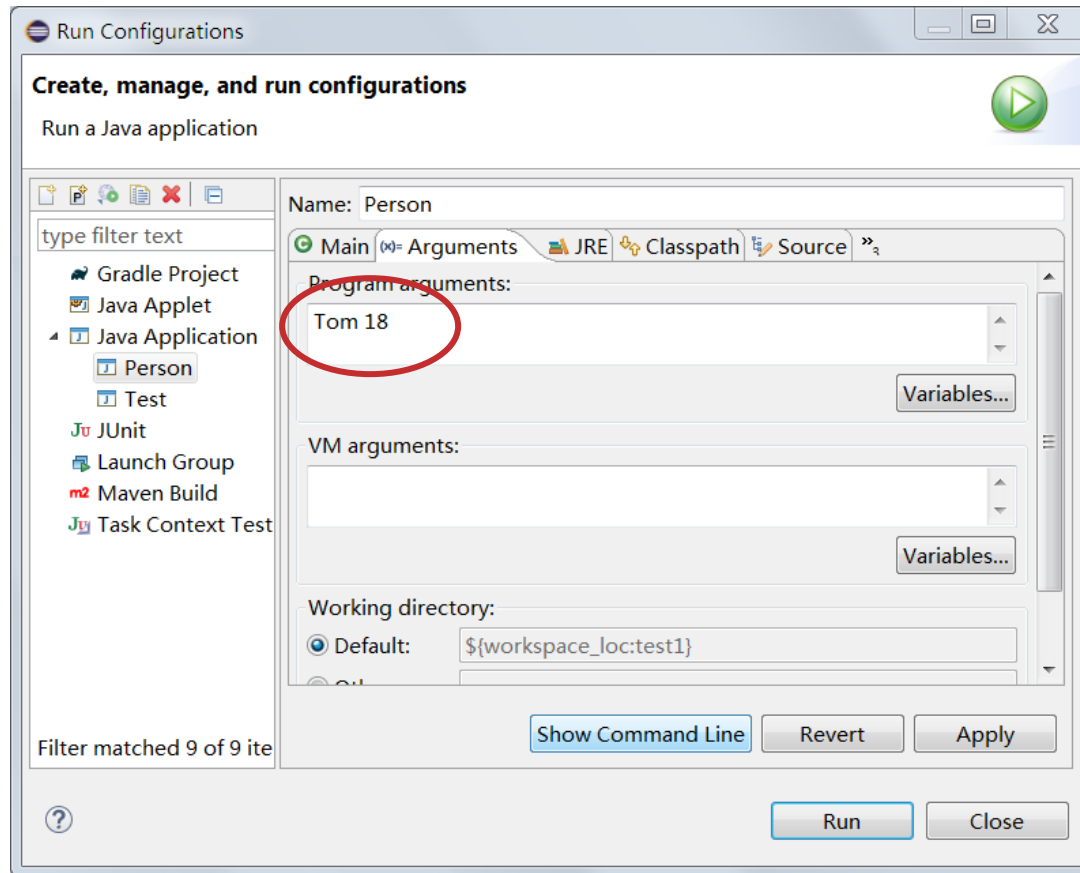
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- Two ways to avoid hard-coding
 - `String[] args`
 - `Scanner`



String[] args

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Scanner

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```
Scanner sc = new Scanner(System.in);
```



hint:

```
package cn.edu.seu.java;  
  
import java.util.Scanner;  
  
public class Person {
```



Lab Work 2

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- A simple version of ATM
 - Single user
 - Deposit / Withdrawal / Query Balance
 - Using Scanner to get user request and amount of money
 - An user interface like this:

```
Please select your transaction:  
1: Deposit  
2: Withdrawal  
3: Query Balance
```

- Try NOT to write all the codes in main()!!



Self-teaching

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- Javadoc
 - What is Javadoc?
 - How to add comments in program for making a Javadoc?
 - How to generate Javadoc in HTML format?
 - How to search in Javadoc?



Forecast

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- OO Concepts
- Class and Objects
 - Package
 - Field
 - Method
 - Main method
 - Object
 - Construct and Initialization
 - Access Control